Andreas Tzitzikas

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Education

University of Maryland, College Park, MD — Expected May 2027
Master of Science in Computer Engineering — **GPA: 4.00**
Bachelor of Science in Computer Engineering (GPA: 3.88) — Expected May 2026
Relevant Coursework: Digital Computer Design (ENEE646), Digital CMOS VLSI Design (ENEE640), Microprocessors (ENEE440), Computer Organization (ENEE350), Digital Circuits Lab (ENEE245), PCB Design (ENEE459A), Operating Systems (ENEE447), Reverse Eng. & HW Security (ENEE459B)

Projects

Dual-Host S2-LP Radio Shield | KiCad, RF Design, Multi-Platform Firmware — Aug 2025 – Dec 2025

- Designed dual-host development board for **S2-LP** sub-1 GHz transceiver with **STM32** and **RP2350** compatibility, implementing complete RF circuit with matching network and crystal oscillator for optimal radio performance.
- Designed a novel "**Shadow Mode**" debugging architecture allowing the **STM32** to snoop **SPI** traffic between the **RP2350** and **S2-LP** radio, facilitating real-time protocol analysis without bus contention.
- Created comprehensive firmware suite for both platforms including SPI drivers, radio configuration, packet transmission, and bus monitoring capabilities with detailed technical documentation.

STM32G4 Bare-Metal Peripheral Integration | ARM Assembly, Embedded C — Jan 2025 – May 2025

- Implemented **bare-metal** firmware with direct register-level programming for **GPIO**, **DAC**, **ADC**, **TIM2** (PWM/PFM), and **SPI2** flash memory drivers on **STM32G4** microcontroller.
- Wrote **bare-metal** drivers (**ARM Assembly**/C) for a resource-constrained **STM32G4**; implemented a multiplexed interrupt handler to service **4 distinct peripherals** via a single hardware IRQ, reducing context-switching overhead.
- Developed comprehensive testing framework ensuring real-time performance, peripheral synchronization, and robust error handling in resource-constrained embedded environment.

Tomasulo Out-of-Order RISC-V CPU | RTL Design, FPGA Implementation, Verification — Nov 2025 - Dec 2025

- Architected complex RTL system using **Tomasulo's Algorithm** with register renaming, **8-entry reservation stations**, and **16-entry reorder buffer** for advanced hardware design.
- Achieved **36% higher operating frequency** (**69.7MHz** vs 51.2MHz) and **40-150% IPC improvement** compared to in-order implementation through hardware optimization and performance analysis.
- Implemented sophisticated branch prediction, speculative execution, and precise exception handling ensuring robust hardware behavior across complex computational workloads.
- Synthesized complex digital system with **1,652 standard cells** achieving **zero DRC violations** and comprehensive functional verification for reliable hardware operation.

Polynomial Evaluation Accelerator | SystemVerilog, Digital Design — Fall 2025

- Architected dataflow-based accelerator with 5 finite state machines managing FIFO handshaking and resource sharing across 9 processing modules for high-throughput computation.
- Designed dual-port coefficient memory with auto-padding logic enabling efficient batch processing of polynomial evaluations with minimal memory access latency.
- Achieved 100% functional coverage through comprehensive testbench development with 15+ complex test cases validated against reference C implementation.

Experience

Embedded Systems Intern, Alchemity – College Park, MD — Jun 2025 – Aug 2025

- Accelerated validation of a hybrid solid oxide fuel cell reactor by developing a full-stack automation suite (**Rust**, **Python**) to parallelize testing, converting multi-day manual experiments into overnight runs to significantly increase throughput.
- Engineered real-time **STM32** firmware to precisely control material synthesis within modular reactors, featuring non-blocking motor/relay drivers and **SPI** peripheral management.
- Built end-to-end control interfaces including desktop GUIs, embedded display drivers, and CLI tools to streamline workflows between hardware and software teams.

Skills

Languages & HDLs: SystemVerilog, Verilog, C/C++, ARM Assembly, Python
Hardware Design: Digital Logic Design, PCB Design, Microcontroller Programming, FPGA Implementation, VLSI Design
Embedded Systems: STM32 Bare-Metal Firmware, Real-Time Control, SPI, I2C, UART, GPIO, ADC/DAC, PWM
EDA Tools: KiCad, Xilinx Vivado, Icarus Verilog, Verible, GTKWave, LPKF Prototyping, Altium Designer
Verification: Functional Verification, Testbench Development, Static Timing Analysis, Linux